Alexander Reinthal

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Senior cyber security practitioner and data engineer switching careers to mitigate loss of control risks through AI control research.

Research Projects

Out of Control Arena *Sept 2026 – present*
Developed a control arena setting to compare AI's capability in cyber defense compared to offense.

- Ported BashArena to a VM setting (Proxmox)
- Extended monitors to perform agentic forensics on victim machines

Aletheia's Quest: Lie detection Hackathon *July 2026 – Aug 2026*
Developed a hybrid lie detector with EleutherAI that generalized over 27 different datasets, 3 different model families and many different types of lies.

- Contributed to new datasets and validating data generation, labeling of data and writing the blog post.
- Awarded 2nd place on the validation set.

Can lying probes flag a Chinese model's deceptive answers? *Jan 2026 – June 2026*
Found that some Chinese models think they are honest when answering politically censored questions.

- Extension of ARENA 7.0 weekend hackathon, 1st place. Supported by Bluedot Impact rapid grant.

About Emergent Misalignment *June 2026 – present*
Found that emergent misalignment (EM) rate is negatively correlated with the capability benchmarks GSM8K (Pearson -0.98) and MMLU (Pearson -0.94).

- Extension of ARENA 7.0 capstone, 3rd place.

Changing the prompt for instrumental convergence evals *May 2026 – June 2026*
Removing pressure phrases increases misalignment rate for capability prompts (He et al. 2025) but decreases it for propensity prompts (Hopman et al. 2026) for Gemini 3.1 Pro Preview.

Detecting Piecewise Cyber Espionage in Model APIs *Nov 2025 – Nov 2025*
Proof of concept for detecting malicious activity across multiple agent sessions.

- Research continued through SPAR by David Williams-King (ERA) and Linh Le (Mila). Apart Research sprint, 4th place of 641 submissions.

Experience

Career Transition Grantee *May 2026 – present*
Independent

- Supported by Coefficient Giving's Career Transition grant for reducing global catastrophic risks. Currently researching lie detection with a broader interest in AI control and scalable oversight agendas.

Research fellow *Sept 2026 – Dec 2026*
SPAR

- Researching Lie detection under Walter Laurito and Kieron Kretschmar (Cadenza Labs).

Alum *Sept 2026 – Sept 2026*
AI Security Bootcamp

- Studied an intense curriculum across AI Safety, AI security and Technical AI governence.

Contractor *Mar 2026 – present*
Bluedot Impact/AI Security Bootcamp

- Interviewed applicants for AI Security bootcamp
- Facilitator and research mentor for the Technical AI Safety course and project.

Alum *Jan 2026 – Feb 2026*
ARENA 7.0

- Five-week intensive curriculum on transformer internals, mechanistic interpretability, RLHF, evals, JAX, and the Inspect framework.

Forward-deployed Engineer / DevSecOps / Tech Lead / Data Scientist *Gothenburg, Sweden*
Knowit Solutions Cocreate *Feb 2022 – present*

- Lead secure deployment of data platforms across multiple clients, some under protective security contracts (Säkerhetsskyddslagen 2018:585).
- Pioneered LLM evals at the company. Implemented an OAuth-authenticated MCP gateway to secure agentic workflows.

Cyber Security Analyst *Gothenburg, Sweden*
NTT Security *Apr 2019 – Jan 2022*

- Wrote calibrated cyber incidents reports based on available evidence. Deployed an ML classifier that reduced analysis time of malicious domains from 2 hours to 10 minutes per analyst per shift.

Python Software Engineer *Gothenburg, Sweden*
Ericsson *June 2017 – Oct 2018*

- First job where I led a team building a log-parsing and analytics tool that got funded and grew into a dedicated engineering team.

Publications

Data Modeling for Predicting Software Exploits Nov 2018
Peer-reviewed paper predicting which CVEs become exploited in the wild. Adopted by Recorded Future's Vulnerability Intelligence product.
Reinthal, A., Filippakis, E., Almgren, M.. 10.1007/978-3-030-03638-6_21 (Springer LNCS: Nordic Conference on Secure IT Systems)

Open Source and Personal Projects

Casually Jailbreaking Gemini 2.5 Flash *2026 – 2026*
Jailbroke Gemini 2.5 Flash with a few-shot in-context continuation attack, eliciting harmful recommendations.

Open Source Contributions
github.com/reinthal

- Inspect AI PR #3987 Fixed Gemini 3.1 Pro + OpenRouter + Reasoning issue

- Dagster PR #24188 (ML pipeline orchestration) and DLT Hub PR #594 (ingestion reliability)

Skills

Machine Learning & AI: Inspect, PyTorch, JAX, linear probes, SFT, RL*

Programming Languages: Nix, Python, JavaScript, SQL, Terraform

Platforms & Tools: Tinker, Modal, Weights & Biases, Spark, Docker, Kubernetes, AWS, Azure

Certifications: Bluedot Impact — Technical AI Safety (2026-03); AGI Strategy (2025-10)

Education

Chalmers University of Technology *Gothenburg, Sweden*
MS in Engineering Physics *Sept 2016 – June 2018*

- Master thesis got published and put into production at Recorded Future.
- Switched from computer science undergraduate to engineering physics graduate program and performed better.

Gothenburg University *Gothenburg, Sweden*
BS in Computer Science *Sept 2013 – June 2016*

- Thesis on finding common dense subgraphs using linear programming has 4 citations.